The maximum static frictional force $f_s^{max} = \mu_s N = \mu_s mg$

$$f_s^{max} = 0.8 \times 2 \times 9.8 = 15.68\,N$$

Therefore to move the object the external force should be greater than maximum static friction.

$$F_{ext} > 15.68\,N$$

EXAMPLE 3.18

Consider an object of mass 50 kg at rest on the floor. A Force of 5 N is applied on the object but it does not move. What is the frictional force that acts on the object?

Solution

When the object is at rest, the external force and the static frictional force are equal and opposite.

The magnitudes of these two forces are equal, $f_s = F_{ext}$

Therefore, the static frictional force acting on the object is

$$f_s = 5N.$$

The direction of this frictional force is opposite to the direction of F_{ext}.

EXAMPLE 3.19

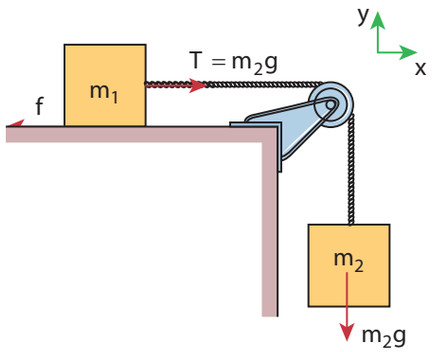
Two bodies of masses 7 kg and 5 kg are connected by a light string passing over a smooth pulley at the edge of the table as shown in the figure. The coefficient of static friction between the surfaces (body and table) is 0.9. Will the mass m_1 = 7 kg on the surface move? If not what value of m_2 should be used so that mass 7 kg begins to slide on the table?

Solution

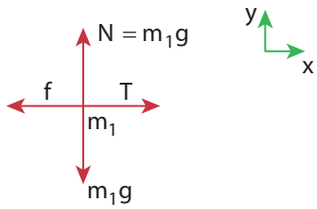
As shown in the figure, there are four forces acting on the mass m_1

a) Downward gravitational force along the negative y-axis (m_1g)
b) Upward normal force along the positive y axis (N)
c) Tension force due to mass m_2 along the positive x axis
d) Frictional force along the negative x axis

Since the mass m_1 has no vertical motion, $m_1g = N$



Free body diagram for mass m_1



To determine whether the mass m_1 moves on the surface, calculate the maximum static friction exerted by the table on the mass m_1. If the tension on the mass m_1 is equal to or greater than this maximum static friction, the object will move.

$$f_s^{max} = \mu_s N = \mu_s m_1 g$$

$$f_s^{max} = 0.9 \times 7 \times 9.8 = 61.74\ N$$

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